A3-PS019-TC049

Personal-Data Discovery and Privacy Evidence Inventory

Team Equinox

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AIORI Hackathon 3

October 13, 2026

Problem Description

PS019 - Personal-Data Discovery and Privacy Evidence Inventory

Personal and sensitive data is often scattered across multiple digital platforms, making it difficult to locate, classify, and monitor. Manual identification of such information is time consuming and prone to errors, increasing the risk of accidental exposure and improper data handling. We have thought for an automated, secure, and centralized solution that discovers personal data, highlights potential privacy risks, and maintains an organized inventory of privacy evidence.

Background

We need an efficient and automated approach to discover personal data, classify sensitive information, identify potential privacy risks, and maintain a structured inventory of evidence.

Keypoints -

- **Growing data volume** : Personal information is spread across multiple digital sources.
- **Limited visibility** : It is difficult to know where sensitive information is stored.
- **Manual effort** : Reviewing files and records individually takes time and can lead to errors.
- **Lack of organized evidence** : Privacy findings may not be documented consistently.
- **Need for optimization** : Automated discovery, classification, and evidence reporting can make privacy reviews faster and more systematic.

Literature & RFC Review

1. NIST Privacy Framework

The NIST Privacy Framework provides a structured, risk-based approach to identifying and managing privacy risks. Its inventory and mapping guidance is relevant to discovering personal data and understanding how it is processed.

2. RFC 6973 — Privacy Considerations for Internet Protocols

RFC 6973 provides guidance for analyzing privacy threats, data collection, identifiability, and possible mitigation measures in internet systems.

3. GDPR — Article 30: Records of Processing Activities

Article 30 of the General Data Protection Regulation describes requirements for maintaining records of personal-data processing for organizations to which the relevant obligations apply. It provides a useful reference for structuring data inventories and documenting how personal information is processed.

4. Microsoft Presidio — Open-Source Data Protection Tool

Microsoft Presidio is an open-source framework for detecting and analyzing personally identifiable information (PII) in text and other supported data formats.

Proposed Solution

We propose to develop a centralized, privacy-focused tool that automatically discovers personal and sensitive information across selected digital files and data sources, classifies the identified information, and generates a structured Privacy Evidence Inventory.

Key Features:

- **Automated Data Discovery :** Identify personal information in supported files and data sources.
- **PII Detection and Classification :** Categorize detected information by type and sensitivity.
- **Privacy Risk Identification :** Flag potentially exposed or improperly stored sensitive data.
- **Evidence Inventory :** Maintain structured records of findings, including source, category, timestamp, and risk level.
- **Report Generation :** Generate clear reports with actionable recommendations.
- **Privacy and Security :** Protect sensitive findings through controlled access and data minimization.

Optimization Proposed

Our proposed system aims to optimize personal-data discovery and privacy assessment by replacing repetitive manual tasks with an automated, centralized, and privacy-focused workflow. The optimizations focus on reducing manual effort, improving consistency, organizing evidence, and making privacy risks easier to identify and address.

Before vs. After Comparison :

Aspect	Before (Existing Approach)	After (Proposed Solution)
Data Discovery	Files checked manually	Automated scanning of supported files
PII Detection	Depends on manual review	Automated detection using patterns and rules
Data Classification	Inconsistent categorization	Standardized categories for sensitive data
Risk Identification	Risks may be overlooked	Potential risks flagged for review
Evidence Management	Findings stored separately	Centralized privacy evidence inventory
Reporting	Reports prepared manually	Automated report generation
Time and Effort	Time-consuming and repetitive	Faster review with reduced manual effort
Privacy Protection	Sensitive data may appear in reports	Masked findings and restricted access

Solution Architecture & Design

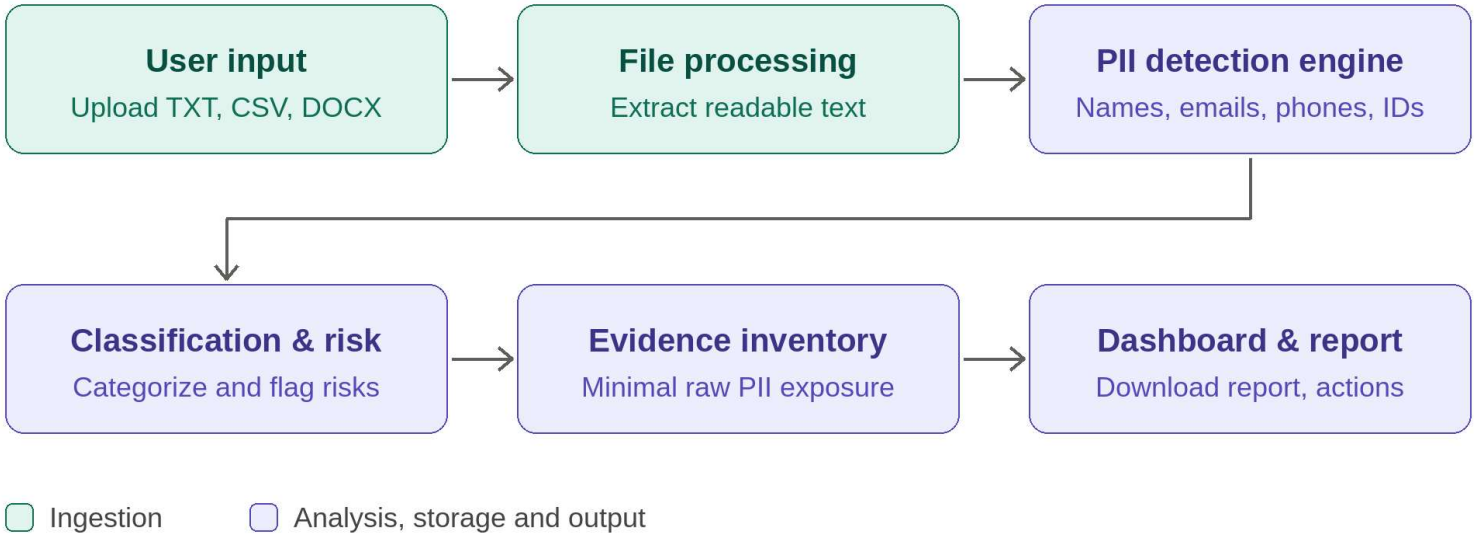
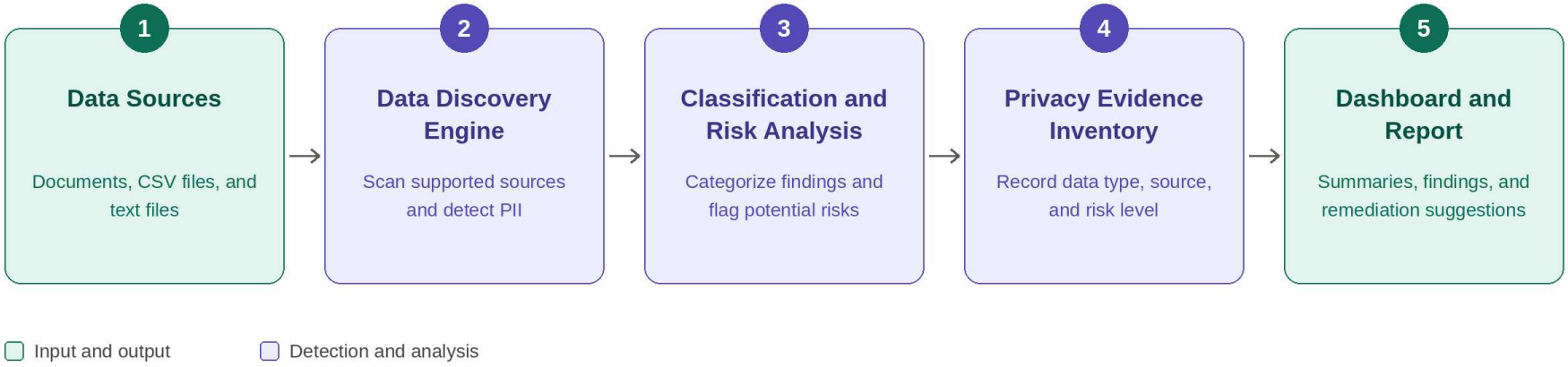


Figure: Solution Design

Data Flow Diagram



Deployment Timeline

We will follow an iterative approach consisting of planning, development, testing, feedback, and refinement. The system will be developed module by module, with each component tested before integration. The final prototype will be evaluated for functionality, usability, detection quality, and privacy protection.

Phase	Milestone	Activities
Phase 1	Planning and Design	Finalize requirements, workflow, and system architecture.
Phase 2	Core Development	Build file upload, text extraction, and preprocessing modules.
Phase 3	PII Detection	Implement detection and classification of personal information.
Phase 4	Inventory and Reporting	Develop the evidence inventory, risk categorization, and report generation.
Phase 5	Testing and Optimization	Test with sample datasets, fix errors, and improve detection quality.
Phase 6	Deployment and Demonstration	Deploy the prototype locally or on a suitable platform and demonstrate the complete workflow.

Bibliography

[1] https://www.nist.gov/privacy-framework/privacy-framework

[2] https://www.rfc-editor.org/rfc/rfc6973.html

[3] https://eur-lex.europa.eu/TodayOJ/index.html

[4] https://github.com/data-privacy-stack/presidio

Thank You!

- TEAM EQUINOX